

CSE 202 Midterm 2 Part (b)

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I. MULTIPLE CHOICE QUESTIONS (40 POINTS)

1. Given the following data $a = (8-9-4-3-2)$ as a queue. What will be the queue after the following operations?

```
a.enqueue(4)
a.enqueue(9)
a.dequeue()
a.enqueue(3)
a.dequeue()
```

- a) 8-9-2-4-9-3
- b) 4-3-2-4-3-8
- c) 9-4-3-8-9-2
- d) 4-3-2-4-9-3
- e) 3-9-4-4-3-8

2. What will be the time complexity of enqueue and dequeue if we implement it with circular linked list?

- a) $\mathcal{O}(1)$, $\mathcal{O}(N)$
- b) $\mathcal{O}(N)$, $\mathcal{O}(1)$
- c) $\mathcal{O}(1)$, $\mathcal{O}(1)$
- d) $\mathcal{O}(\log N)$, $\mathcal{O}(1)$
- e) $\mathcal{O}(\log N)$, $\mathcal{O}(\log N)$

3. If the size of the queue is N (linked list implementation), how many elements can we store in that queue?

- a) N
- b) $N - 1$
- c) $N + 1$
- d) $N - 2$
- e) $N + 2$

4. In the game of Darts, if we have 3 different circles to shot to (5, 10, 20), which states will be traversed to search for two shots?

- a) 0, 5, 10, 20
- b) 0, 5, 10, 15, 20, 30
- c) 0, 5, 10, 15, 20, 25, 30, 40
- d) 0, 5, 10, 15, 20, 25, 30
- e) 5, 10, 20

5. For array implementation, which of the following code fragment check for a queue with only one element?

- a) `return first + 1 == last;`
- b) `return first == last + 1;`
- c) `return (first + 1) % N == last;`
- d) `return first == (last + 1) % N;`
- e) `return first == last;`

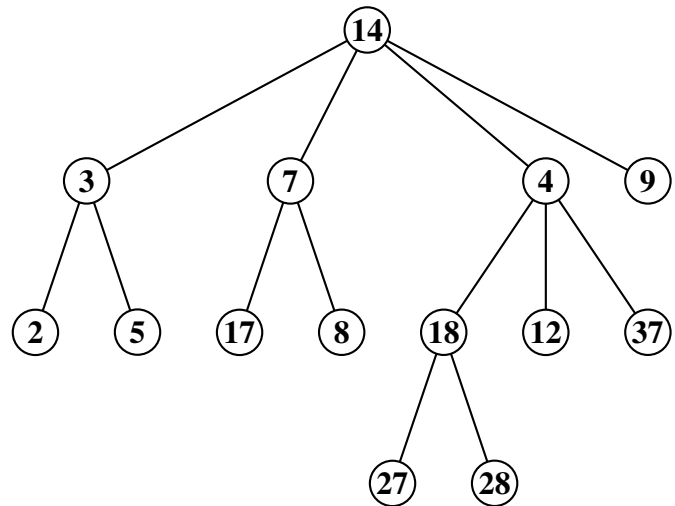
6. For linked list implementation, what are the linked list operations to implement a FIFO (first in, first out) queue?

- a) enqueue: insertFirst, dequeue: removeLast
- b) enqueue: removeLast, dequeue: insertFirst

c) enqueue: insertLast, dequeue: removeFirst

d) enqueue: removeFirst, dequeue: insertLast

e) enqueue: insertFirst, dequeue: removeFirst



7. Given the tree above, what is (i) the root of the tree, (ii) degree of node 7, (iii) depth of the tree?

- a) 7, 3, 3
- b) 14, 2, 2
- c) 9, 4, 3
- d) 14, 3, 4
- e) 14, 2, 4

8. What are the minimum and maximum number of nodes in a binary tree of depth 4?

- a) 4, 12
- b) 5, 31
- c) 3, 7
- d) 4, 15
- e) 6, 10

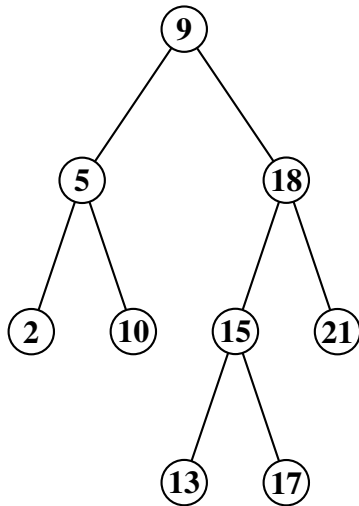
9. If we want to obtain a fully inbalanced binary search tree from numbers 1 to 7, what is the correct order of insertion of those numbers to the binary search tree?

- a) 4, 5, 6, 2, 1, 3, 7
- b) 1, 2, 3, 4, 5, 6, 7
- c) 4, 2, 6, 1, 3, 5, 7
- d) 3, 5, 7, 2, 6, 4, 1
- e) 4, 2, 1, 3, 5, 6, 7

10. What is the best, worst time complexity for insertion in a fully balanced binary search tree?

- a) $\mathcal{O}(1)$, $\mathcal{O}(\log N)$
- b) $\mathcal{O}(1)$, $\mathcal{O}(N)$
- c) $\mathcal{O}(1)$, $\mathcal{O}(1)$

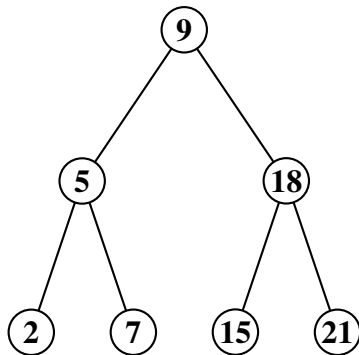
- d) $\mathcal{O}(\log N)$, $\mathcal{O}(\log N)$
 e) $\mathcal{O}(\log N)$, $\mathcal{O}(N)$



11. Given the tree above, which node does not satisfy the binary search tree property?

- a) 18
 b) 10
 c) 13
 d) 21
 e) 17

12. For the tree given below, where can we insert 14?



- a) Right child of 7
 b) Left child of 7
 c) Right child of 15
 d) Left child of 15
 e) Left child of 21

13. Given the tree in Question 11, which of the following shows the inorder traversal of the tree?

- a) 2, 10, 5, 13, 17, 15, 21, 18, 9
 b) 2, 5, 10, 9, 13, 15, 17, 18, 21
 c) 9, 5, 2, 10, 18, 15, 13, 17, 21
 d) 9, 5, 18, 2, 10, 15, 21, 13, 17
 e) 2, 10, 13, 17, 21, 5, 15, 18, 9

14. What is the maximum number of values that can be stored in a node in a B+ tree with $d=3$?

- a) 3
 b) 4
 c) 5
 d) 6
 e) 7

15. Which of the following numbers may be used as a hash table size?

- a) 105
 b) 14
 c) 29
 d) 38
 e) 64

16. Given an hash table of size 9, which of the following shows the resulting hash table after inserting 18, 23, 66, 34?

- a) [-, -, 23, 66, 18, -, 34, -, -]
 b) [18, -, 66, 34, -, 23, -, -, -]
 c) [18, -, -, -, -, 23, 34, 66, -]
 d) [18, -, -, 66, -, 23, -, 34, -]
 e) [-, 23, 66, 18, -, 34, -, -, -]

17. Which of the following methods is a possible collision resolution strategy with linked lists?

- a) Separate Chaining
 b) Linear Probing
 c) Quadratic Probing
 d) Double Hashing
 e) Hash Function

18. What is the best, worst time complexity for inserting in hash table?

- a) $\mathcal{O}(1)$, $\mathcal{O}(\log N)$
 b) $\mathcal{O}(1)$, $\mathcal{O}(N)$
 c) $\mathcal{O}(1)$, $\mathcal{O}(1)$
 d) $\mathcal{O}(\log N)$, $\mathcal{O}(\log N)$
 e) $\mathcal{O}(\log N)$, $\mathcal{O}(N)$

19. Given an hash table of size 17, what is the suggested size of the hash table after rehashing?

- a) 18
 b) 34
 c) 37
 d) 105
 e) 101

20. Which of the following index(es) can be used for equality queries ($x = 10$)?

- a) Hash Index
 b) None
 c) Hash Index, Tree Index
 d) Tree Index
 e) Tree Index, sometimes Hash Index